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Subject:

Recommendation for Dr.-Ing. Barış İrhan

Dr. Barış İrhan, born on 10.04.1977 in Ankara (Turkey), obtained his M.S. in Civil Engineering at Middle East Technical University, Ankara, Turkey in 2002. Subsequently he obtained his M.Sc. at the University of Stuttgart (COMMAS) in 2003. From 2011-2014 he was working under my supervision on his Ph.D. Thesis: "High velocity impact and fragmentation of concrete: Numerical simulation". Dr. İrhan is currently working at the University of Stuttgart, Institute of Construction Materials, on further improvements of our finite element codes.

During his stay at our Institute he was working in the framework of two DFG research projects: "Dynamic fracture of concrete" and "Impact and fire". He was responsible for development of the 3D finite element code for multi body dynamic simulations with contact, fragmentation and thermo-mechanics. The theoretical framework was standard finite elements with total Lagrangian formulation. As a constitutive law the rate and temperature dependent microplane model was used with Green-Lagrange strain tensor and co-rotational Cauchy stress tensor. To model contact and impact problems erosion strategy was employed. Moreover, as a part of the code Dr. İrhan also developed a number of useful pre- and post-processing tools. As results of his work the finite element code CIF was developed that was at our institute successfully used in several research projects.

In the numerical fields of research Dr. İrhan has excellent skills. He was very much interested in optimal solving of complex tasks through constructive proposals and he made a decisive contribution in their realization in very efficient way. Moreover, as a person he was willing to help and cooperate with the members of our research group.

In short, Dr. İrhan is an outstanding experienced person in the field of computational mechanics with very strong commitment towards learning everything that is relevant to the field of computational mechanics and programming in different computer languages. He has more than on one occasion shown outstanding initiative and creativity in solving different very complex numerical problems.

With kind regards,

(Prof. Dr.-Ing. Josko Ozbolt)